

Genome comparisons

Sjoerd Seekles

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- Msc in Leiden – Biotech
  - Internship Han de Winde on ISS12 *Pseudumonas putida*
- PhD in Leiden – Arthur Ram
  - Heterogeneity in spores of food spoiling fungi focus *Aspergillus niger*
- Post-doctoral in Lausanne / Geneva
  - Mate choice preferences of wild *Schizosaccharomyces pombe* (fission yeast) strains

Switzerland

- Group prof. Sophie Martin
- Fission yeast (*Schizosaccharomyces pombe*) mating
  - molecular machinery guiding cell-cell fusion, pheromone signaling and mate choice
  - Compare wild isolates of fission yeast
- Fungal genetics & genomics, bioinformatics (genome comparisons)



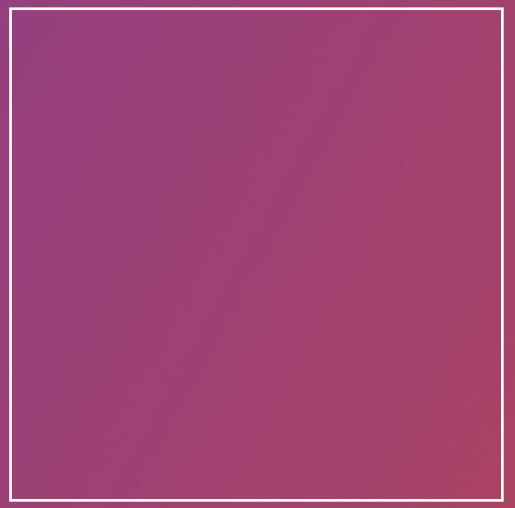
Presentation in two parts

- Part 1
  - from raw sequence reads > genome
- Break
- Part 2
  - From genome > annotated genomes and simple genome comparisons
- Break
- Part 3
  - Simple genome comparisons > complex genome comparisons

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Attendees: 0

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What words do you associate with genomes?

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What words do you associate with genomes?



Time left: -

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Votes: 0

Genomes

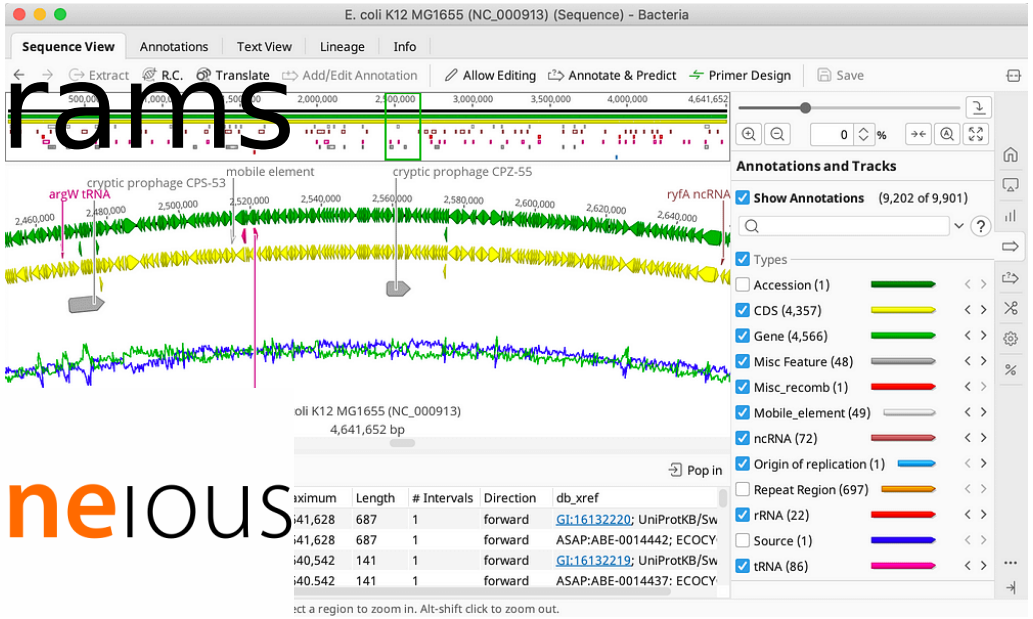
- DNA or RNA
- Eukaryote or Prokaryote
- Genes
- Open reading frame (ORF)
- Coding and non-coding regions
- Genomic variation / mutations
- Chromosomes
- Comparing genomes?

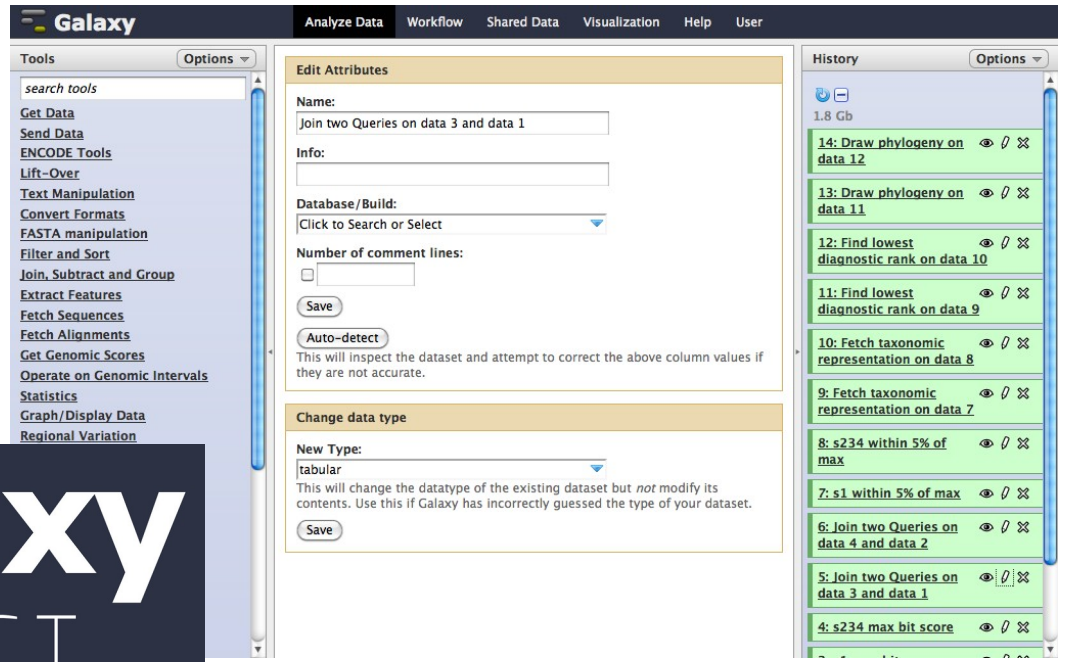


Popular all in one programs

- Geneious
  - User friendly UI
  - Everything in one spot
- Galaxy project
  - Use modules you choose

- All follow the same basic steps
  - Linux based software





https://www.geneious.com/

Enis Afgan, Dannon Baker, Bérénice Batut, Marius van den Beek, Dave Bouvier, Martin Čech, John Chilton, Dave Clements, Nate Coraor, Björn A Grüning, Aysam Guerler, Jennifer Hillman-Jackson, Saskia Hiltemann, Vahid Jalili, Helena Rasche, Nicola Soranzo, Jeremy Goecks, James Taylor, Anton Nekrutenko, Daniel Blankenberg, The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2018 update, *Nucleic Acids Research*, Volume 46, Issue W1, 2 July 2018, Pages W537–W544, https://doi.org/10.1093/nar/gky379

Where to start?

- Know the genomes you are comparing
  - Plants / Fungi / Bacteria / Mammalian etc.
- Research question

- Internet
  - NCBI (Genbank)
- Sequencing



Marta’s UV mutant



Boeke lab

- Marta did UV mutagenesis using a lab strain (yeast)
- Found interesting phenotype of this UV mutant
  - Maybe it suddenly made purple pigment!
- How to know what change in the genome caused this interesting phenotype?
- Sequence the UV mutant to obtain the genome

From sequencing to genome file

- Let’s say Illumina sequencing was performed (short-reads)
- Sequencing will give you raw reads
  - .fastq files
- First step is some quality control and cleanup
  - Quality check
  - Trimming and quality filtering of reads

How does a Fastq file look like? Its simply DNA reads see below

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Quality Control (QC) on raw sequence reads

- FastQC, why?
- Illumina reads are based on light pulses
  - Generated by a PCR based method
  - Errors



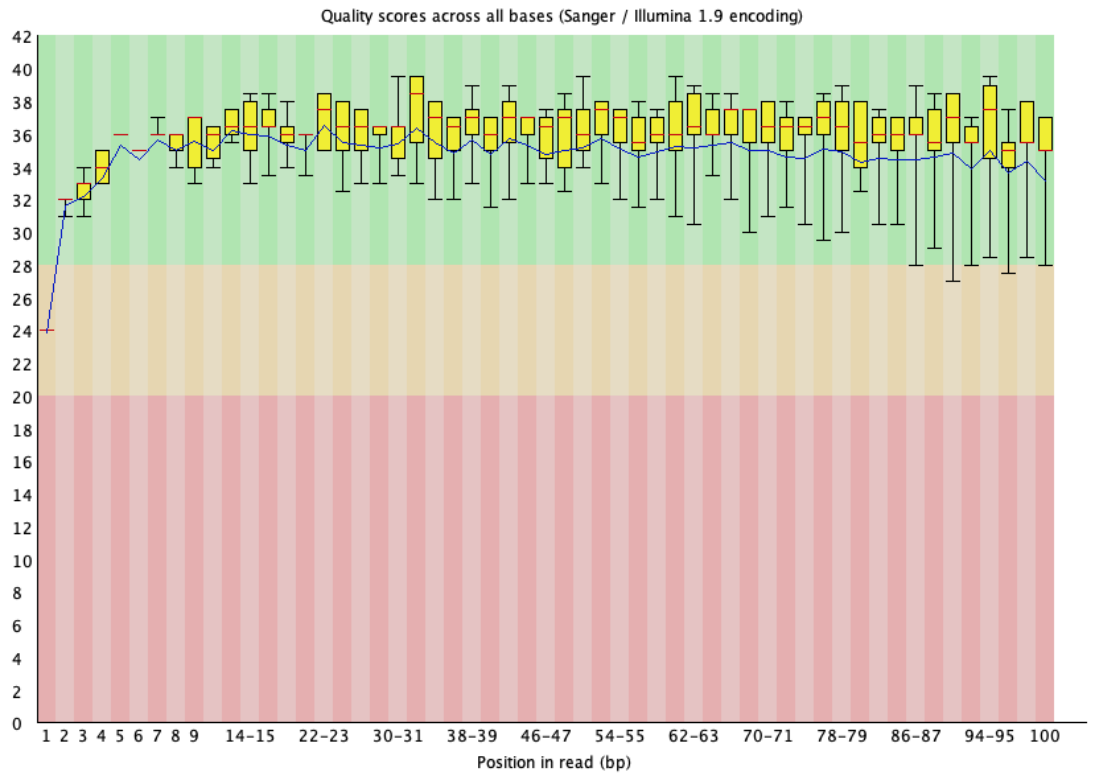
Fastqc –f fastq reads1.fastq reads2.fastq –o outputfolder

https://www.synthego.com/help/sanger-sequencing-quality

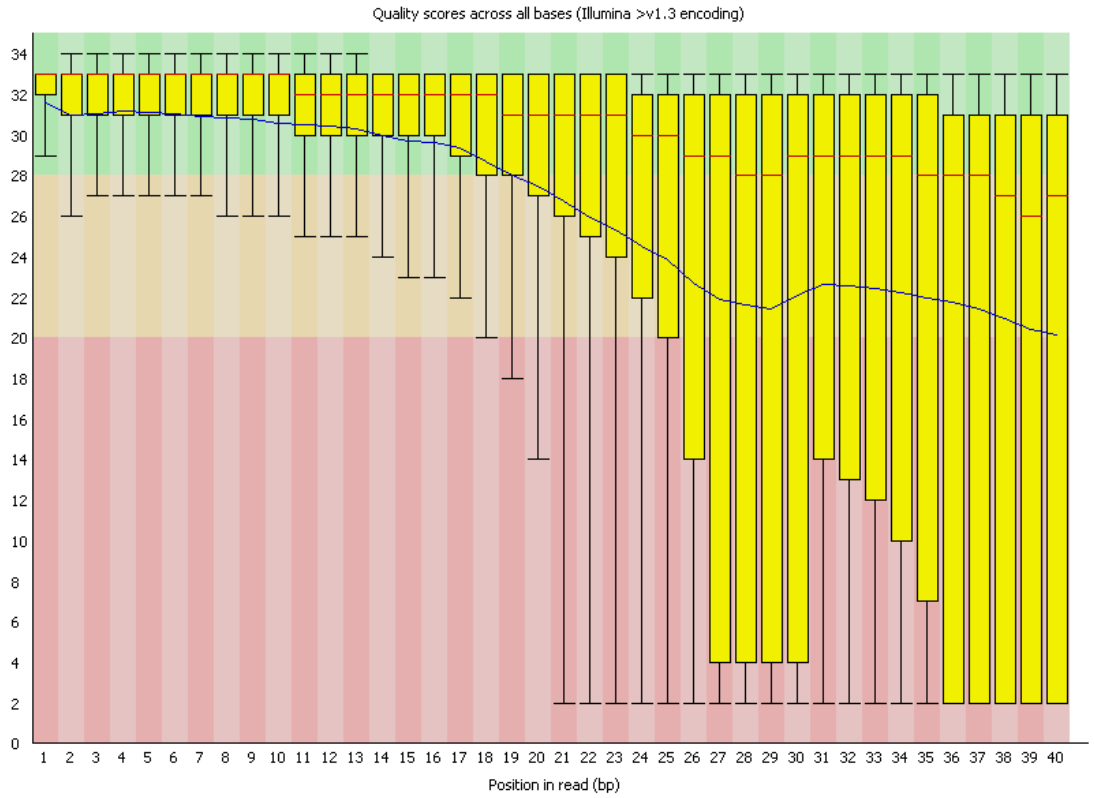
Andrews S. (2010). FastQC: a quality control tool for high throughput sequence data. Available online at: http://www.bioinformatics.babraham.ac.uk/projects/fastqc

FastQC output

Good quality reads

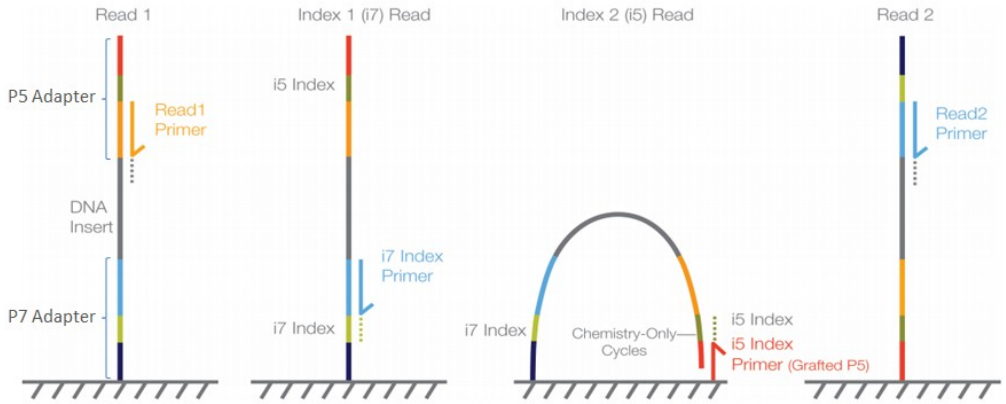


Bad quality reads



Read trimming

- Read trimming is the process of shortening the reads
  - Removal of low-quality reads
  - Removal of known adapter sequences
- Illumina sequencing technology relies on adapter sequences. They have two functions
  - Attachment of your shortened genome to surface
  - Provide template for standard primer
  - Provide clear end point of read



Illumina technologies

- In the sequencing process your genome fragment is flanked by adapter sequences
- During the sequencing process, the adapter on the opposite side of your DNA fragment is sequenced
- These are known sequences at the end of your reads, they need to be trimmed off
  - BBDuk, trimmomatic, cutadapt

Example run BBDuk to remove **adapters**

bbduk.sh in=reads.fq out=clean.fq ref=adapters.fa ktrim=r

Example run BBDuk to remove **low quality reads**

bbduk.sh in=reads.fq out=clean.fq maq=20

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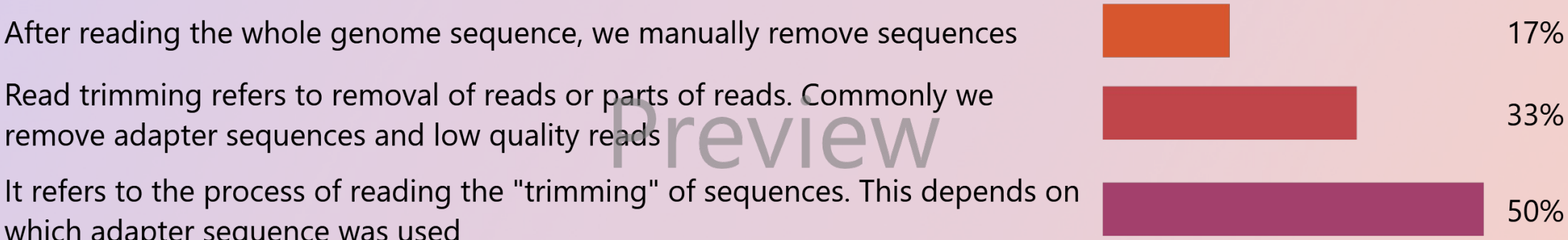
What do we mean by read trimming?

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What do we mean by read trimming?



Preview

From sequencing to genome file

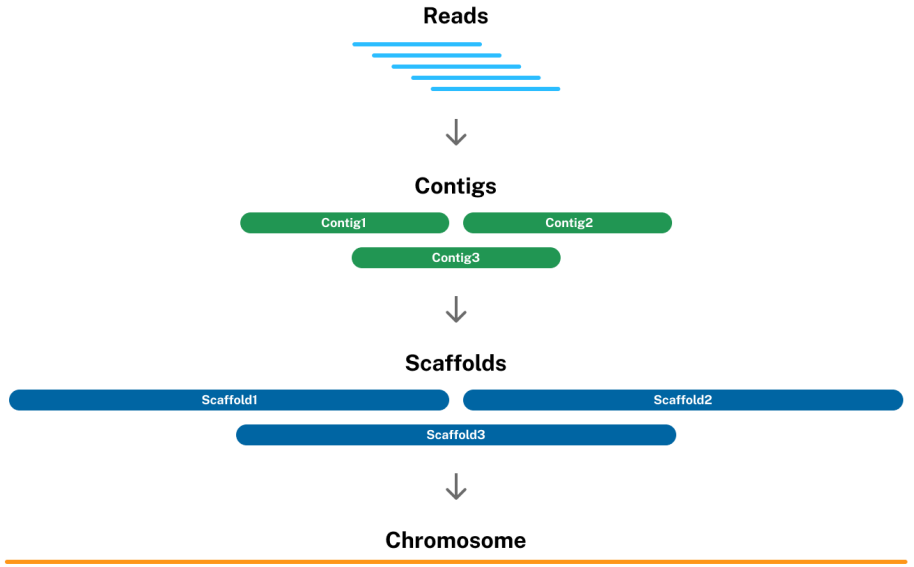
- Let’s say Illumina sequencing was performed (short-reads)
- Sequencing will give you raw reads
  - .fastq files
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Reads are cleaned up!
Now process them for your purpose

For example:
make a genome file using the reads
“Genome assembly”

How does this “genome assembly” work?

- Choose genome assembly method
  - DNA reference mapper
  - *De novo* reference mapper
- Be aware of type of sequencing reads
  - Short-read (Illumina)
  - Long-read (Nanopore, PacBio)
  - Hybrid assemblers



- After using a genome assembler, you will go from fastq files to “.fasta” files.
  - chromosomes, scaffolds, contigs

How does this “genome assembly” work?

- Assemble (small) reads into one (or several big) pieces of genome
  - De novo assembly is like Lego
  - Reference mapper uses template DNA

- Software choice is often preference, however
  - Match the sequencing type (long reads vs short reads) to the assembler!
  - Is there a reference genome available? Reference mapper might be best

- Software settings to pay attention to
  - Very little, perhaps kmer-length (this is genome size dependent!!)

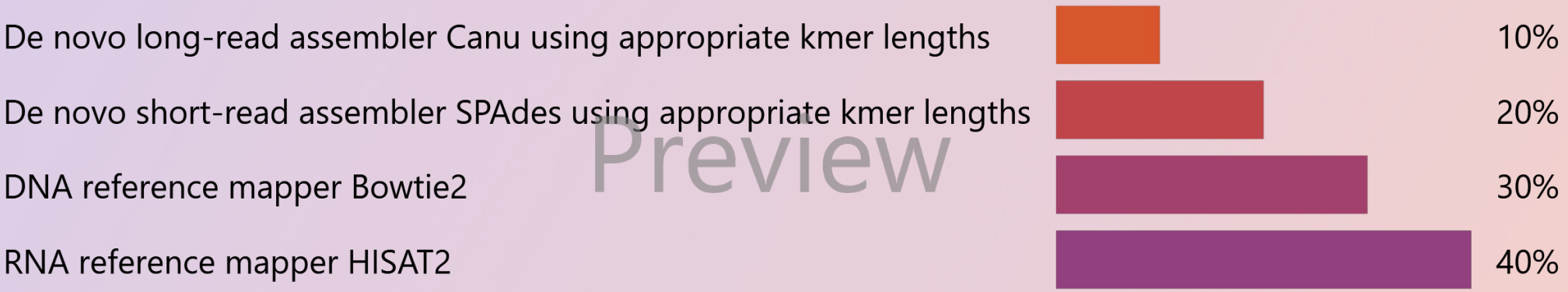
- Examples from SPAdes program, using paired-end Illumina reads:
  - Spades.py –k 21,33,55,77 -1 JB858_1.fastq -2 JB858_2.fastq –o JB858.fasta

Bankevich A, Nurk S, Antipov D, Gurevich AA, Dvorkin M, Kulikov AS, Lesin VM, Nikolenko SI, Pham S, Prjibelski AD, Pyshkin AV, Sirotkin AV, Vyahhi N, Tesler G, Alekseyev MA, Pevzner PA. SPAdes: a new genome assembly algorithm and its applications to single-cell sequencing. J Comput Biol. 2012 May;19(5):455-77. doi: 10.1089/cmb.2012.0021. Epub 2012 Apr 16. PMID: 22506599; PMCID: PMC3342519.

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Marta used Illumina to sequence a yeast mutant, obtained with UV mutagenesis. She wants to have a complete genome, which assembly techniques would be best to use? (multiple options can be correct)

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Preview

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Votes: 0

From genome assembly to annotation

- Proper annotation of genomes is a lot harder
  - Eukaryote vs Prokaryote
  - Transcriptome data helps immensely

- Methods
  - *Ab initio*
  - Evidence-based
- Most recommended is a combination of the two
- Augustus + Braker2 is a great option



Stanke M, Morgenstern B. AUGUSTUS: a web server for gene prediction in eukaryotes that allows user-defined constraints. Nucleic Acids Res. 2005 Jul 1;33(Web Server issue):W465-7. doi: 10.1093/nar/gki458. PMID: 15980513; PMCID: PMC1160219.

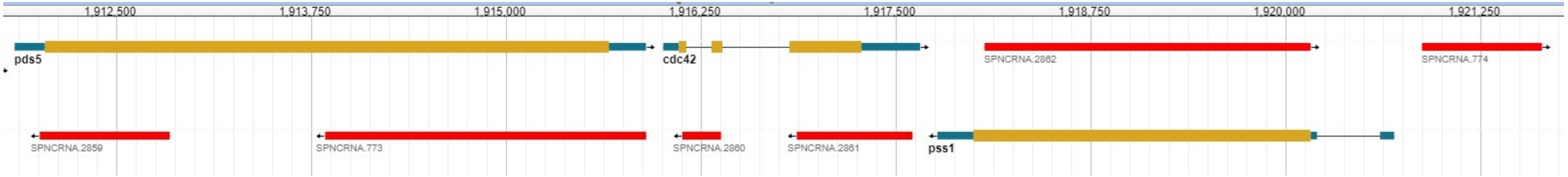
Tomáš Brůna, Katharina J Hoff, Alexandre Lomsadze, Mario Stanke, Mark Borodovsky, BRAKER2: automatic eukaryotic genome annotation with GeneMark-EP+ and AUGUSTUS supported by a protein database, *NAR Genomics and Bioinformatics*, Volume 3, Issue 1, March 2021, lqaa108, https://doi.org/10.1093/nargab/lqaa108

Genome annotation – Augustus + Braker

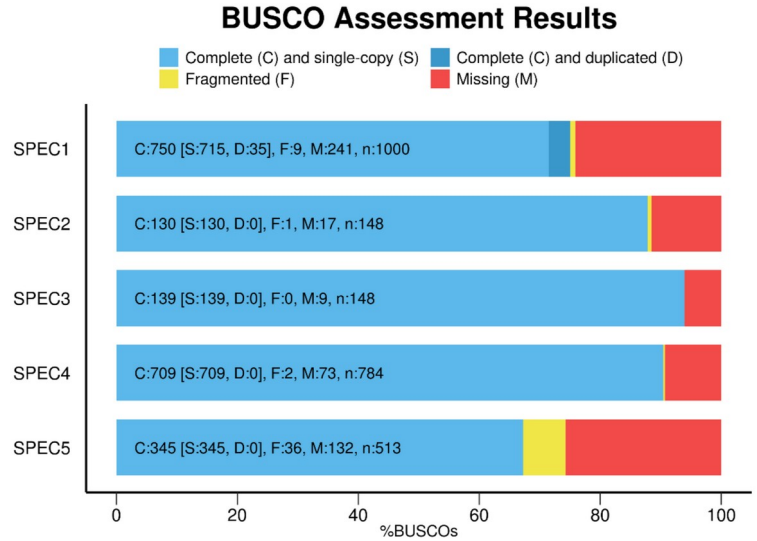
- Mask repeat regions
- Align RNA-seq reads
- Braker2 and Augustus gene prediction
  - Braker generates a set of training genes based on RNA-seq (if any)
  - Augustus, having certain gene prediction rules, gets trained on the Braker dataset
  - Final prediction of genes by Augustus, taking “hints” from RNA-seq
- Here, settings matter a lot more! Use correct species settings!
- Example of getting annotation file (.gff3)
  - Augustus --gff3=on --progress=true --species=schizosaccharomyces_pombe JB858.fasta > JB858.gff3

Quality check annotation

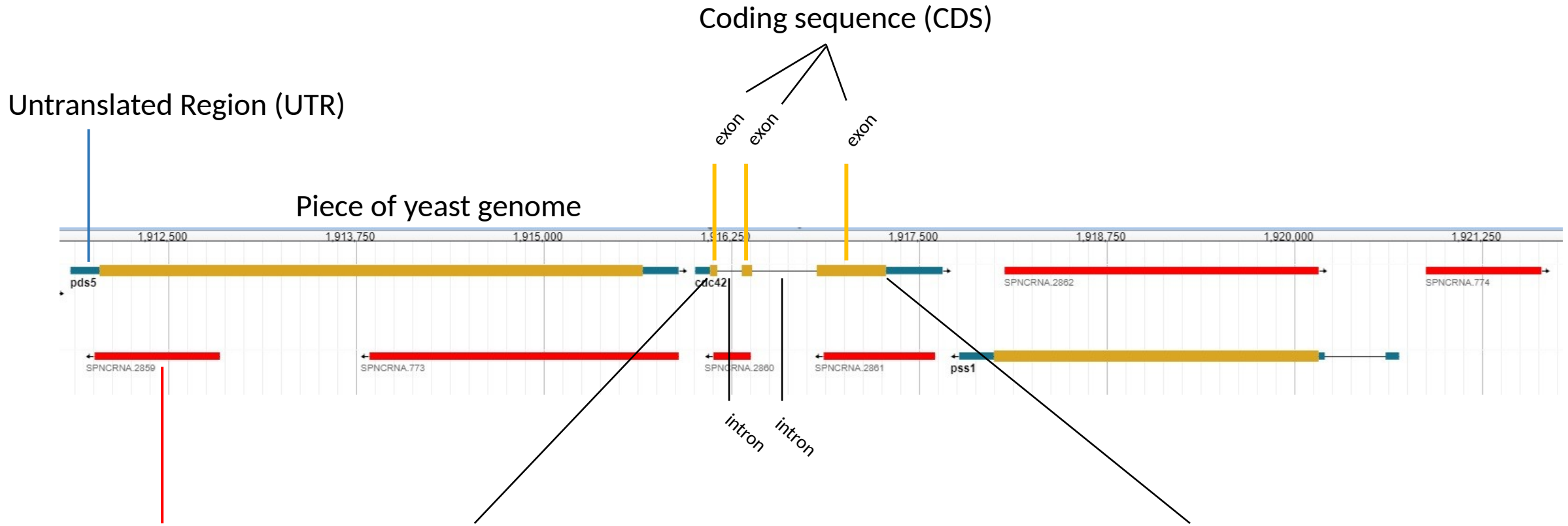
- Check for proper annotation
  - Visualization using program like Integrative Genome Viewer (IGV)



- BUSCO
  - All organisms have a core set of genes essential for live
- BUSCO calculates (depending on organism) how many of the essential genes are annotated in your genome
  - BUSCO score lower than ~90%?
  - Redo annotation, use different settings



Naming conventions eukaryotic genomes



Any given section of genome can be referred to as a “locus”. Multiple sections of genome are “loci”

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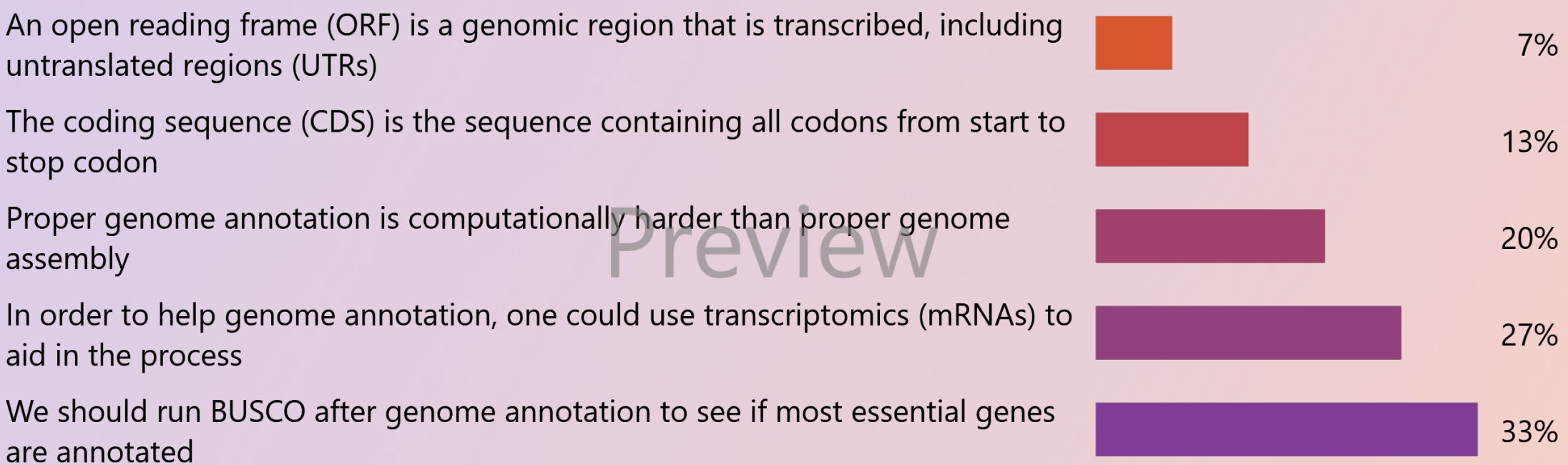
Which of the following statements are correct? (multiple options can be correct)

Time left: -

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Votes: 0

Which of the following statements are correct? (multiple options can be correct)



Preview



Break

We need some coffee!

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Which of the following terms do you know? And do you know their differences?

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Votes: 0

Which of the following terms do you know? And do you know their differences?



Preview

Time left: -

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Votes: 0

Genome comparisons terminology

- Single nucleotide polymorphisms (SNPs)
  - Single nucleotide changes that represent at least 1% of the species' population
  - Historically used for wild strains or known alterations common within a species
  - For example, your blood type

- Single nucleotide variants (SNVs)
  - Single nucleotide changes, period. Marta's lab-made UV mutant contains SNVs but would officially not be considered SNPs.

- Genomic variants
  - All genomic changes, this includes larger deletions, insertions and genome re-arrangements. This also includes SNVs and is therefore the safest but least precise term.

Task completed! Or…?

- Marta has a yeast genome that is fully annotated!
- But… does this answer her research question?

- Turns out she is not interested in the genome specifically; she wants to know which mutation in this UV mutant yeast strain causes the interesting phenotype observed in the lab

GWAS

Let’s compare some yeast genomes



How is my mutant strain different from wild type?

- When comparing genomes you need
  - Reference genome
  - Sequenced your strain of interest
- Start with raw sequence files from your mutant yeast strain
- This time you use reference mapper like Bowtie2
  - Reference is the genome of your parental strain / normal lab strain

- So, you downloaded the genome of fission yeast from internet (preferably the same strain!!) and use Bowtie2 for mapping of your raw reads from the sequence machine

Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. Nat Methods. 2012 Mar 4;9(4):357-9. doi: 10.1038/nmeth.1923. PMID: 22388286; PMCID: PMC3322381.

Reference mapper to compare genomes

- Genome Wide Association Studies (GWAS)
  - Comparing SNVs

- Map the raw reads of Marta's mutant yeast to the fasta genome file of its parental strain the wild type
  - Bowtie2
  - SAMtools (results in BAM files)

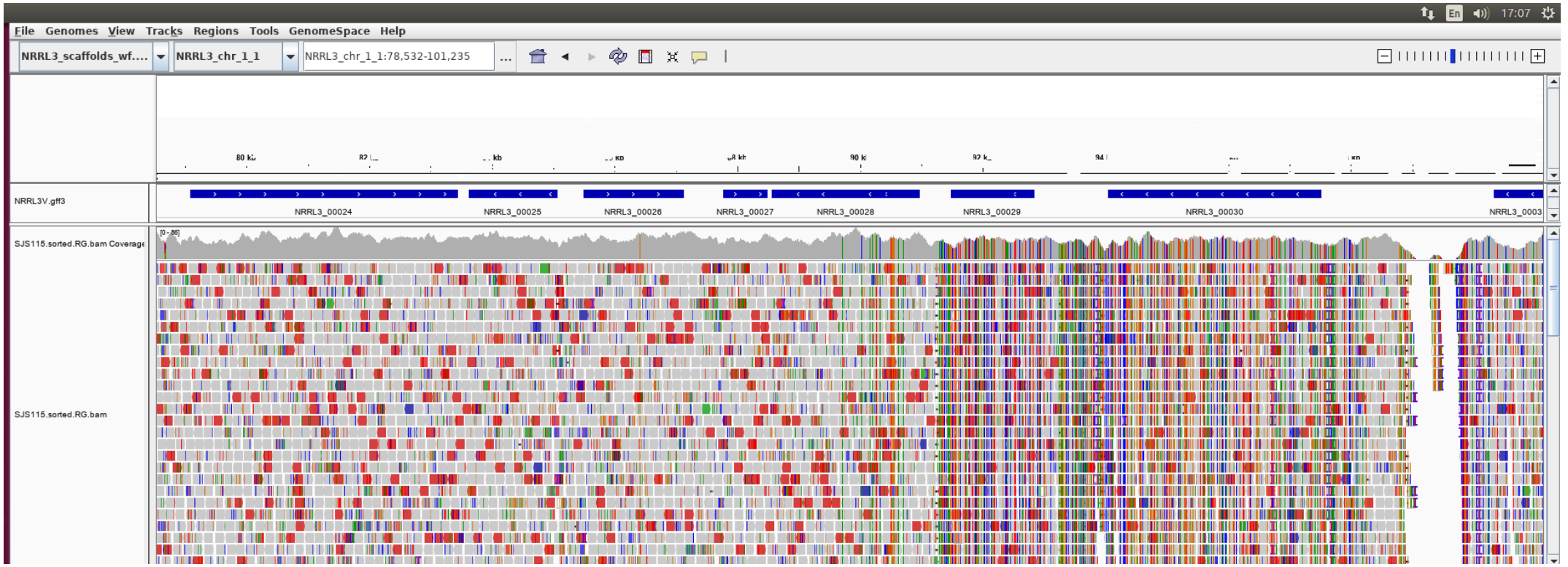
```
bowtie2 -x indexed_genome_name -1 reads_1.fastq -2 reads_2.fastq -S output.sam
samtools view –Sb output.sam output.bam
samtools sort output.bam output.sorted.bam
```

Twelve years of SAMtools and BCFtools
Petr Danecek, James K Bonfield, Jennifer Liddle, John Marshall, Valeriu Ohan, Martin O Pollard, Andrew Whitwham, Thomas Keane, Shane A McCarthy, Robert M Davies, Heng Li
GigaScience, Volume 10, Issue 2, February 2021, giab008, https://doi.org/10.1093/gigascience/giab008

Visualize your mapping

Fastq files are just bases (A,T,C,G). But SAM/BAM files are alignments. These raw reads mapped to the reference genome can be visualized, see example IGV

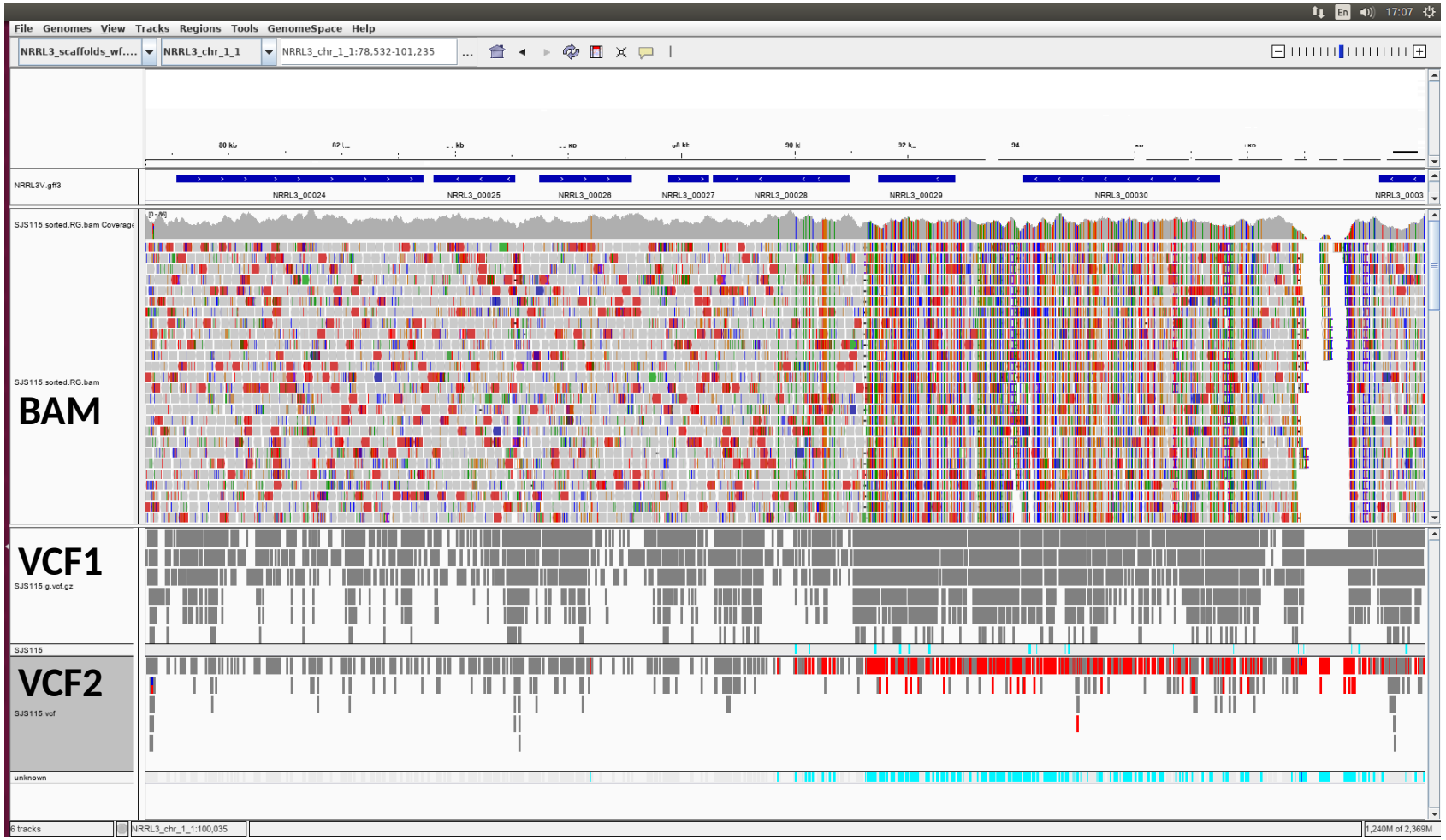
BAM



James T. Robinson, Helga Thorvaldsdóttir, Wendy Winckler, Mitchell Guttman, Eric S. Lander, Gad Getz, Jill P. Mesirov.
Integrative Genomics Viewer. Nature Biotechnology 29, 24–26 (2011). A public access version is also available: PMC3346182.

From BAM file to possible SNVs

- Variant caller
- Many programs, settings are important
  - These programs determine what is called a variant and what not
- VCF1 = FreeBayes
  - Liberal
- VCF2 = GATK Haplotypecaller
  - Strict



https://arxiv.org/abs/1207.3907

Poplin R, Ruano-Rubio V, DePristo MA, Fennell TJ, Carneiro MO, Van der Auwera GA, Kling DE, Gauthier LD, Levy-Moonshine A, Roazen D, Shakir K, Thibault J, Chandran S, Whelan C, Lek M, Gabriel S, Daly MJ, Neale B, MacArthur DG, Banks E. (2017). **Scaling accurate genetic variant discovery to tens of thousands of samples** *bioRxiv*, 201178. DOI: 10.1101/201178

From a file with SNVs (.vcf) to impact

- Once you pinpointed the SNVs, you need to make educated guesses
  - How? Let’s say you have 1000 SNVs, how to determine the most likely candidate?

From a file with SNVs (.vcf) to impact

- Once you pinpointed the SNVs, you need to make educated guesses
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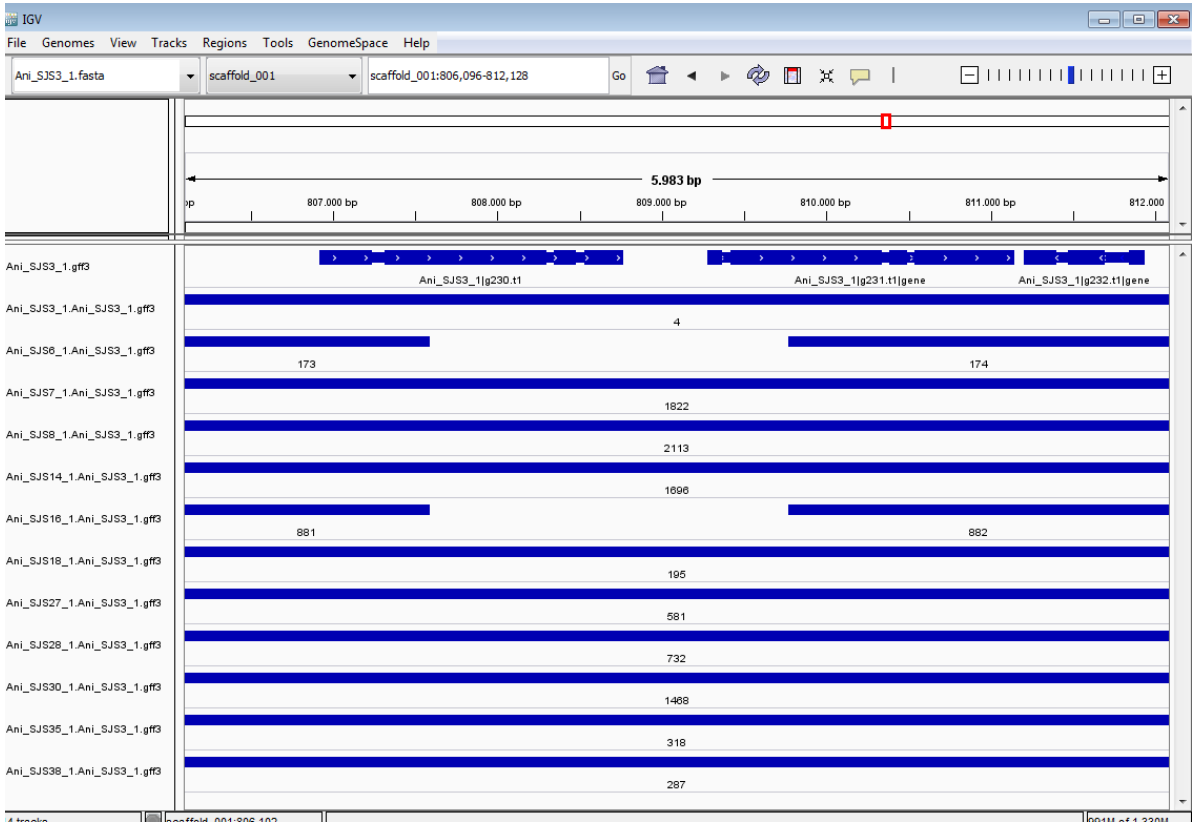
- SNVs inside genes more important than outside genes?
- SNVs causing a frameshift inside gene (deleterious) more important than silent mutations?

- SnpEff
  - Classifies high and low impact SNPs based on annotation file

"A program for annotating and predicting the effects of single nucleotide polymorphisms, SnpEff: SNPs in the genome of Drosophila melanogaster strain w1118; iso-2; iso-3.", Cingolani P, Platts A, Wang le L, Coon M, Nguyen T, Wang L, Land SJ, Lu X, Ruden DM. Fly (Austin). 2012 Apr-Jun;6(2):80-92. PMID: 22728672

Genome comparisons - Advanced

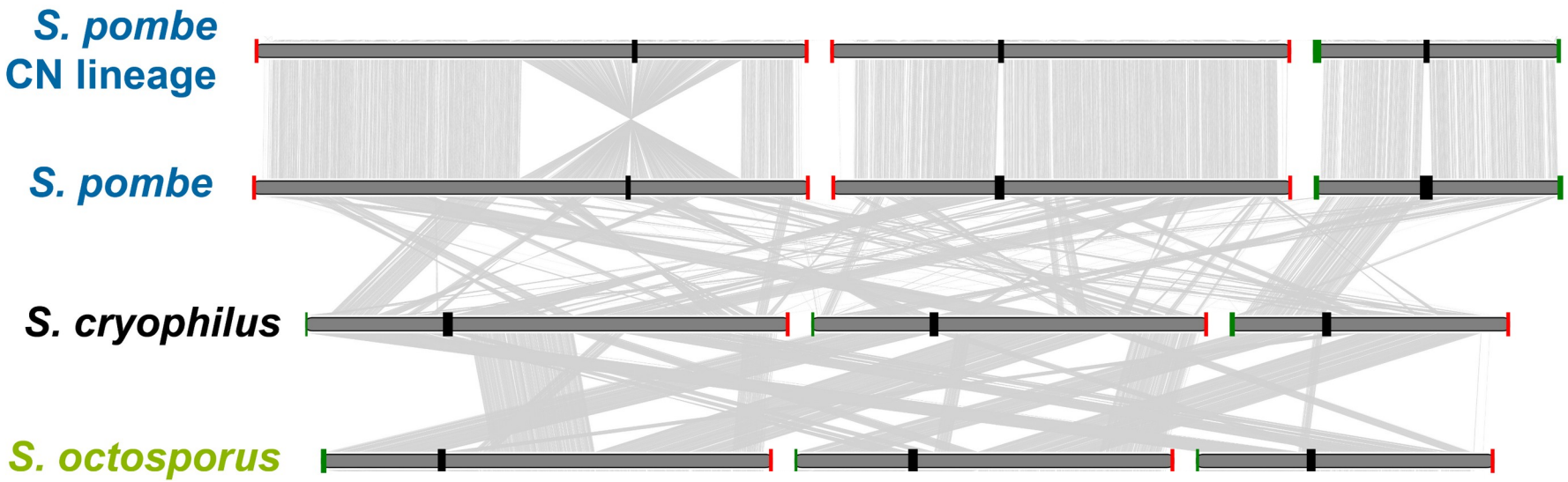
- SNVs are small changes, what about larger genomic changes?
- Genome vs genome mapping
  - MUMmer / NUCmer
- Large, but simple, alignments



"Versatile and open software for comparing large genomes." S. Kurtz, A. Phillippy, A.L. Delcher, M. Smoot, M. Shumway, C. Antonescu, and S.L. Salzberg, *Genome Biology* (2004), 5:R12

Genome synteny plot

- Synteny
  - Co-localization of genetic loci across species
  - Genes and their order are conserved between species
  - Gives us an idea of how related species are and how conserved certain genome arrangements are



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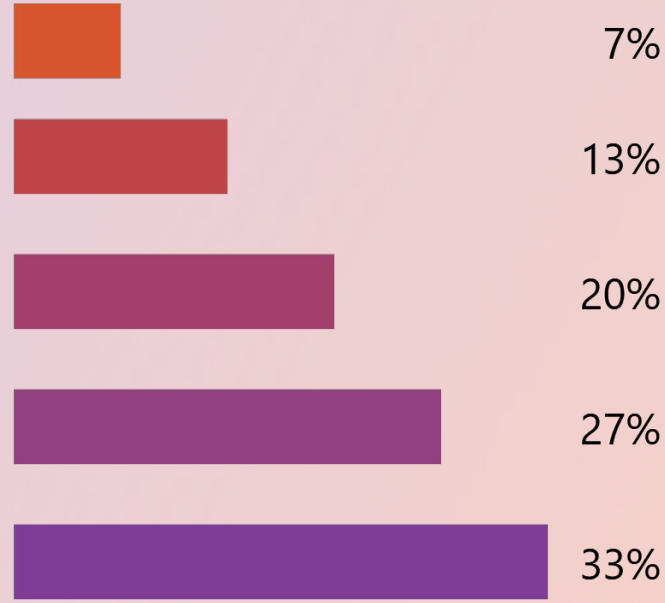
Which of the following statements is true

Time left: -

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Votes: 0

Which of the following statements is true



Preview

Done?

- Marta calculated the SNVs and made an overview in Excel
- She found 10 SNVs, 2 are inside a gene
- According to SnpEff, one is low impact (one amino acid change) and one is high impact (causes early stop codon)

- It seems likely that the mutation causing an early stop codon is the cause!!
- Time to write a paper about it… or?

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Is Marta done? Has she deciphered the causal mutation?

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Votes: 0

Is Marta done? Has she deciphered the causal mutation?



Preview

Time left: -

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Votes: 0

Causality must be experimentally verified!

- We have a likely candidate, but cannot conclude this SNV is the cause of the changed phenotype of Marta’s mutant

- Need to experimentally verify the mutation causes the phenotype by
  - Restoring the mutation in the UV-mutant yeast strain back to wild type
  - Introducing the mutation in the wild-type yeast parental strain

- Most commonly used method nowadays is CRISPR/Cas9 genome editing for targeted change of single bases

Break

We need some coffee!



What about more than 2 strains?

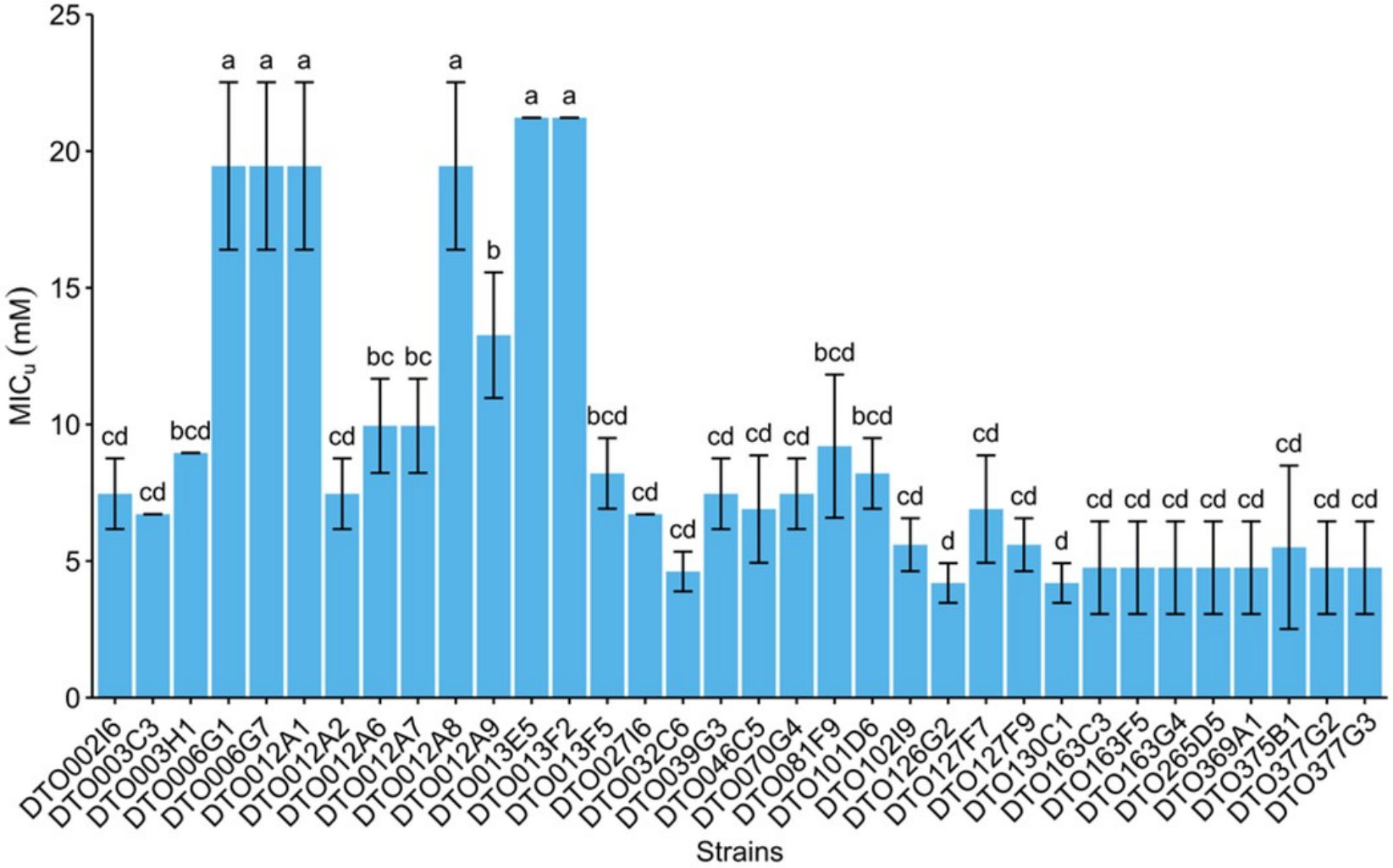
- Filamentous fungus *Penicillium roqueforti*
  - Blue cheese fungus
  - Spoils (non-cheese) dairy products
  - Survive food preservatives such as sorbate

- New research question
  - We have 34 *Penicillium roqueforti* isolates from all over the world
  - Some have high sorbate resistance, others low
  - What gene(s) cause sorbate resistance?



Punt M, Seekles SJ, van Dam JL, de Adelhart Toorop C, Martina RR, Houbraken J, et al. (2022) High sorbic acid resistance of *Penicillium roqueforti* is mediated by the SORBUS gene cluster. PLoS Genet 18(6): e1010086. https://doi.org/10.1371/journal.pgen.1010086

Six strains have high sorbate resistance



Genomes retrieved, but how to proceed?

- What genetic element causes the high sorbate resistance?
- Whole genomes sequencing (Illumina)
- Created fasta files with annotation
  - Fastq > fasta using SPAdes
  - Gene annotation using AUGUSTUS, provided RNA-seq data

- How do we get our genes of interest? Which strains to compare?

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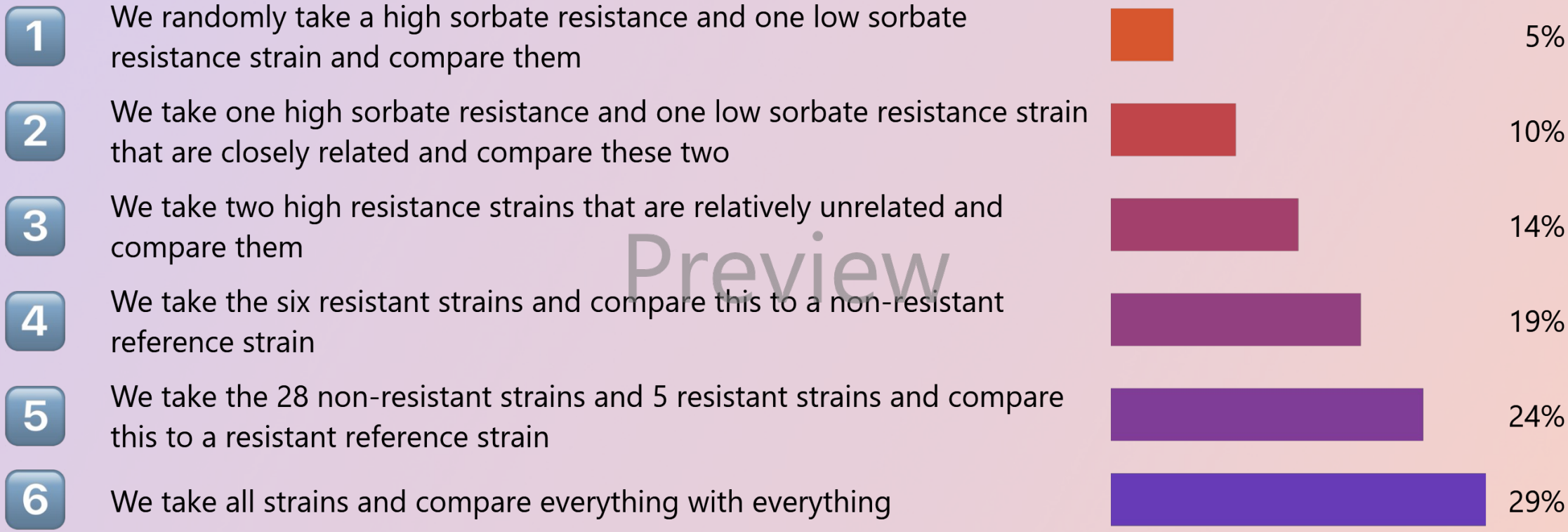
How do we compare this large number of genomes to find our genes of interest? (multiple options can be correct)

Time left: -

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Votes: 0

How do we compare this large number of genomes to find our genes of interest? (multiple options can be correct)



Preview

All options have uses, let’s discuss

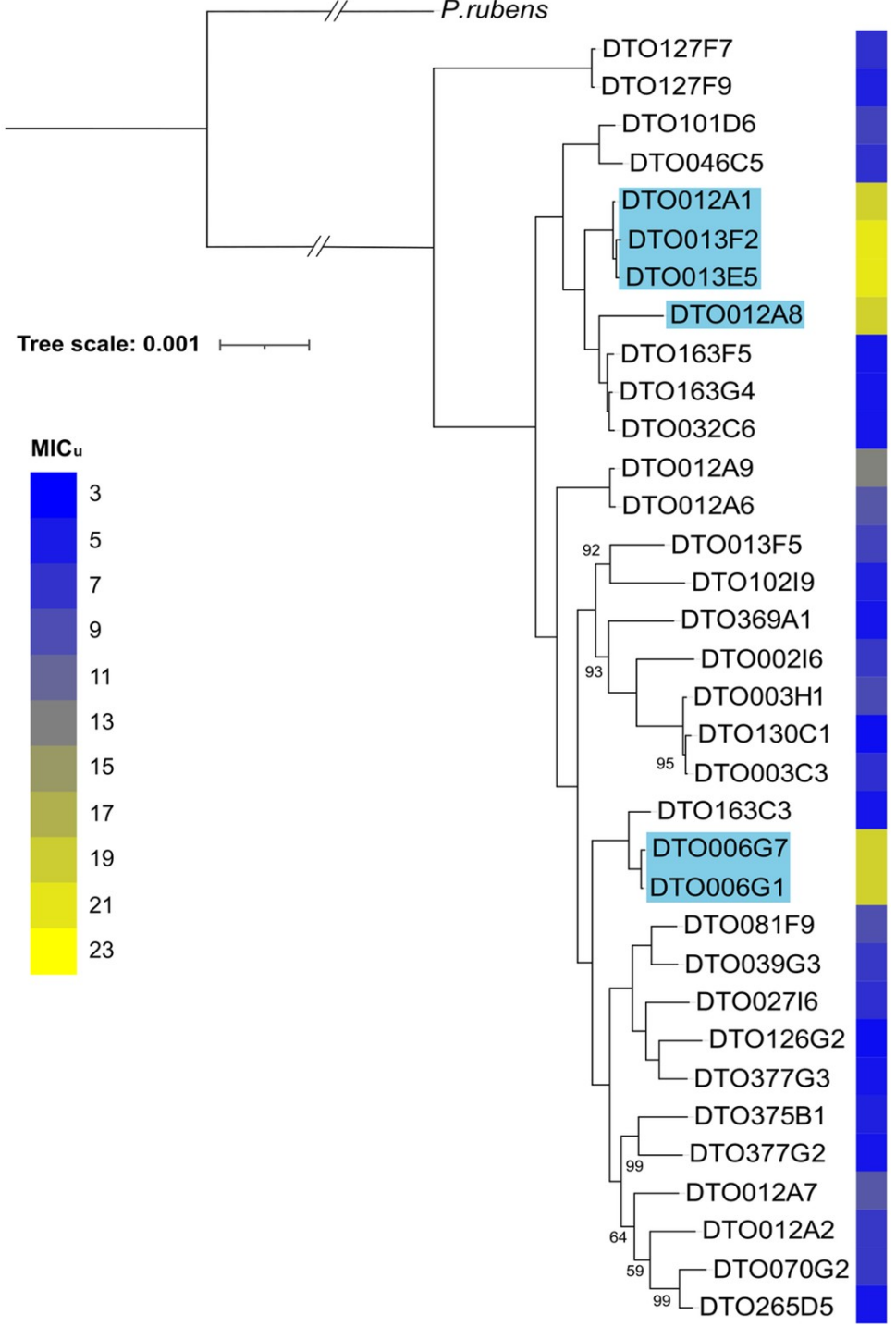
1. Take a single high resistance and low resistance strain to compare
   - This is what we did with Marta.
   - Fast, but might leave you with too many possibilities

2. Take a high and low resistance strain that are closely related
3. Take two high resistance strains that are relatively unrelated

Options 2 & 3 are relatively fast and good approaches, need phylogeny!

Phylogenetic tree

- Which strains are more related?
- 16S / ITS sequencing, MEGA
- Using whole genomes
- Orthofinder
  - Concatenated protein files of orthologues
  - Alignment (MAFFT)
- RaxML
  - From alignments make phylogenetic tree
  - Bootstrap (1000x)
- Tree visualization
- Which strains make sense to compare?
  - Six high resistance strains marked in blue



Emms, D.M., Kelly, S. OrthoFinder: phylogenetic orthology inference for comparative genomics. *Genome Biol* **20**, 238 (2019). https://doi.org/10.1186/s13059-019-1832-y

A. Stamatakis: "RAxML Version 8: A tool for Phylogenetic Analysis and Post-Analysis of Large Phylogenies". In Bioinformatics, 2014, open access.

Comparing multiple strains

4. Take six resistant strains and compare to non-resistant reference
5. Take 28 non-resistant strains and 5 resistant strains and compare to resistant reference

It makes sense to use high resistant strain as reference, why?

Make the individual comparisons 1 by 1 and overlap the data

See which SNV is unique for your reference strain

Computational or time intensive method!

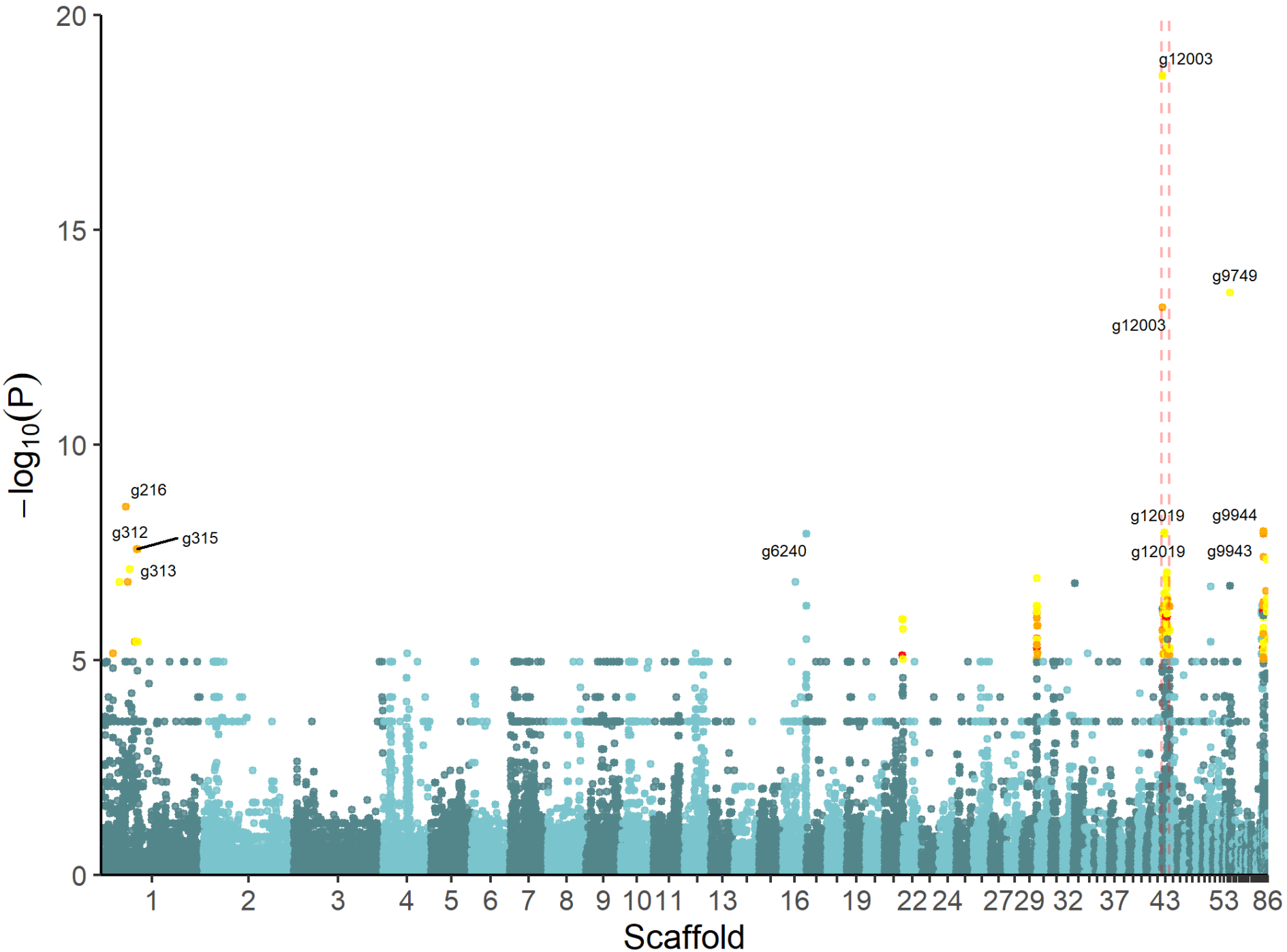
Comparing all vs all

- Why don’t we just compare everything?!

- Computationally demanding
- What are we comparing?
  - Group based on sorbate resistance
  - Phenotype and hypothesis should be leading
    - High resistant strain = reference

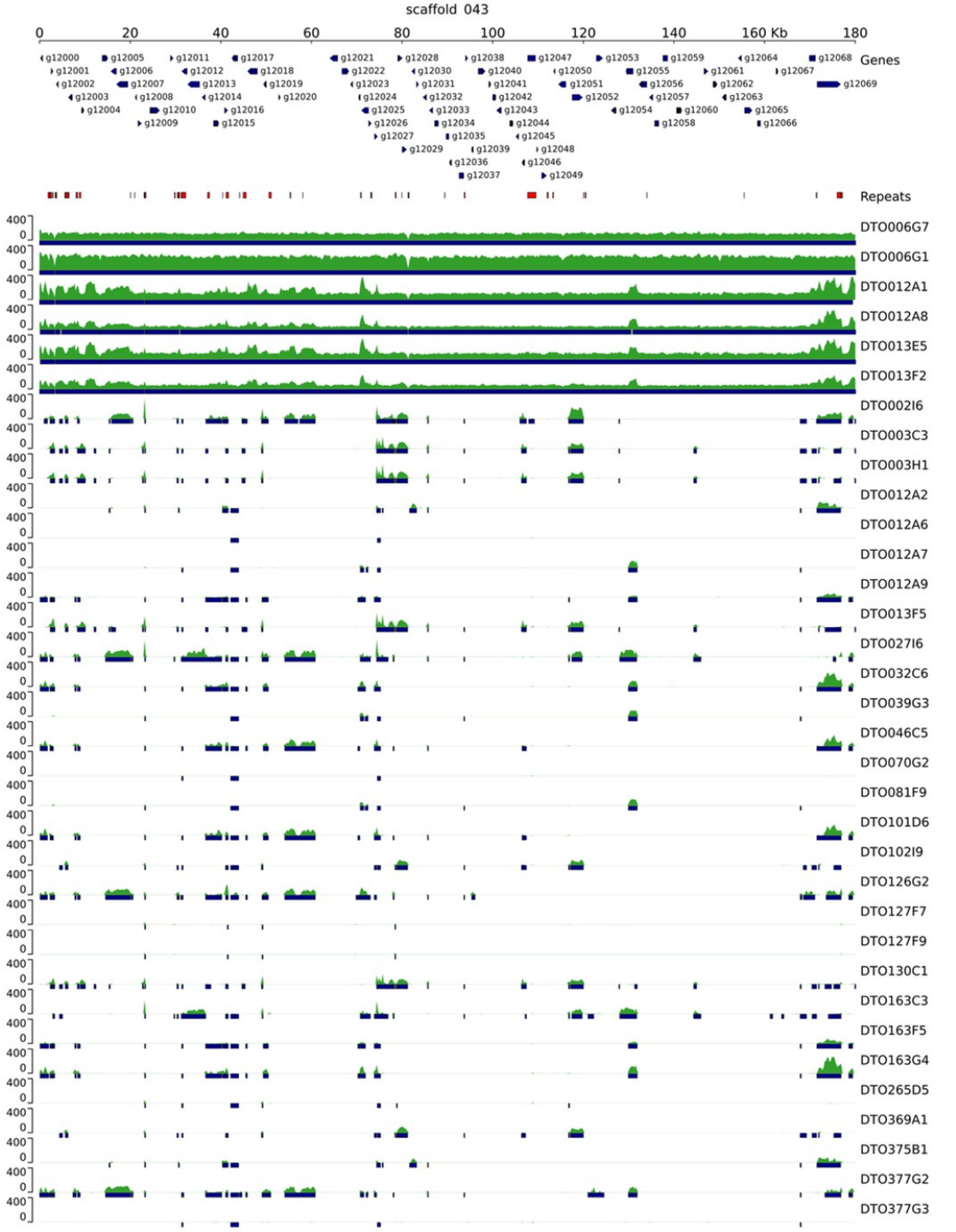
We made all the vcf and nucmer files, now what?!

- How to get an overview of all these SNV comparisons (option 5)?
- Filter based on significance of SNV
- SnpEff
  - Relate SNVs to high impact on genes, combine data in big Excel sheet
- Link SNV presence to phenotype with PLINK
  - SNV more often present in high resistant strain, absent in sensitive strain
- Calculate p-values for each SNV with PLINK
- Commonly used visualization is a Manhattan plot


20
15
10
5
0
−log10(P)
g12003
g9749
g12003
g216
g312
g315
g313
g6240
g12019
g9944
g12019
g9943
1
2
3
4
5
6
7
8
9
10
11
13
16
19
22
24
27
29
32
37
43
53
86
Scaffold

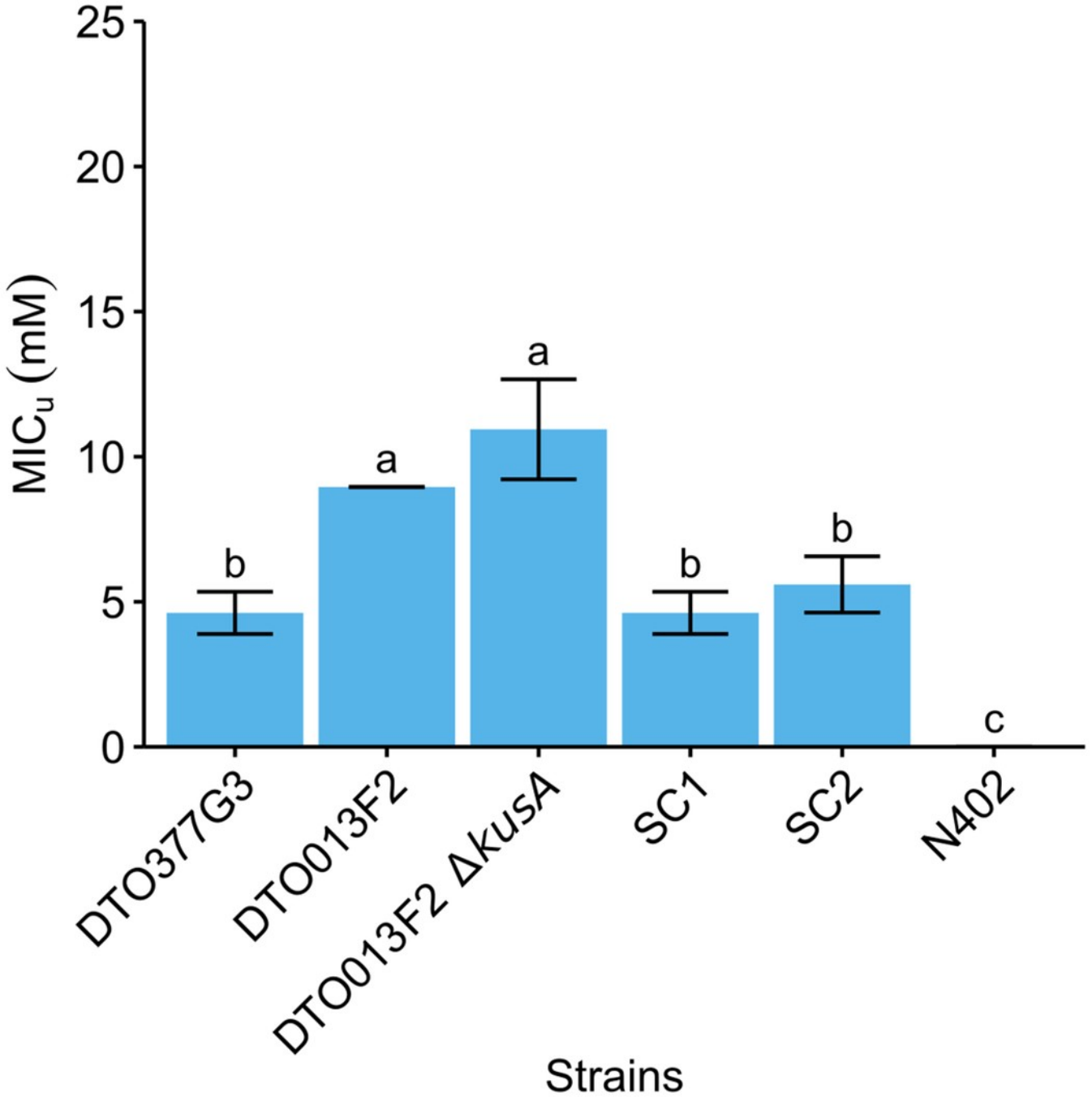
Scaffold 43

- Green = presence
- Present in all sorbate resistant strains
- Absent in the other strains



Test causality

- Scaffold 43 deleted in sorbate resistant strain
- Figure shows that deletion mutants SC1 and SC2 became sorbate sensitive



Bulked segregant analysis and QTL mapping

How to find causal mutation present in one strain with large amount of (unique) SNVs?



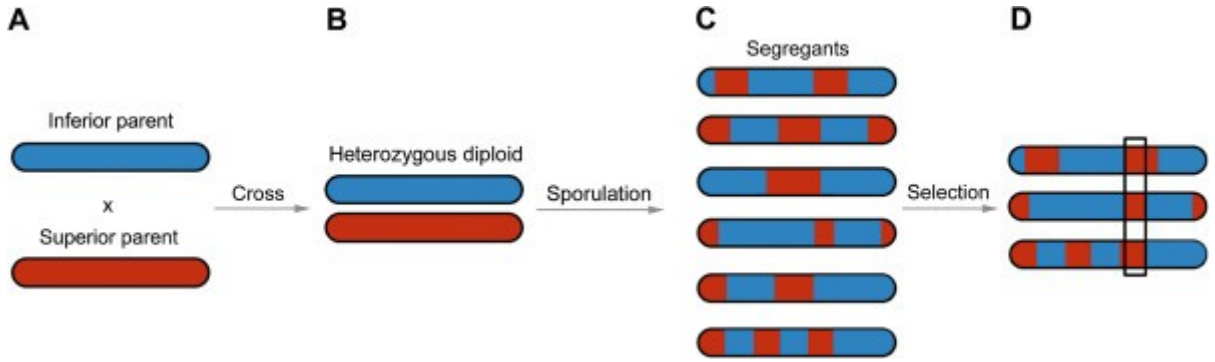
Imagine Marta went overboard on the UV…

- What to do if the UV mutant of Marta doesn’t have 10 SNVs but 1000?
  - In case of interesting (wild) strain it can have 10,000+ SNVs
  - What if these SNVs are all unique? There isn’t any other strain with the same phenotype that we can use for genome comparisons like in the *Penicillium roqueforti* example.

- Now we need to clean up the genome using **bulked segregant analysis**

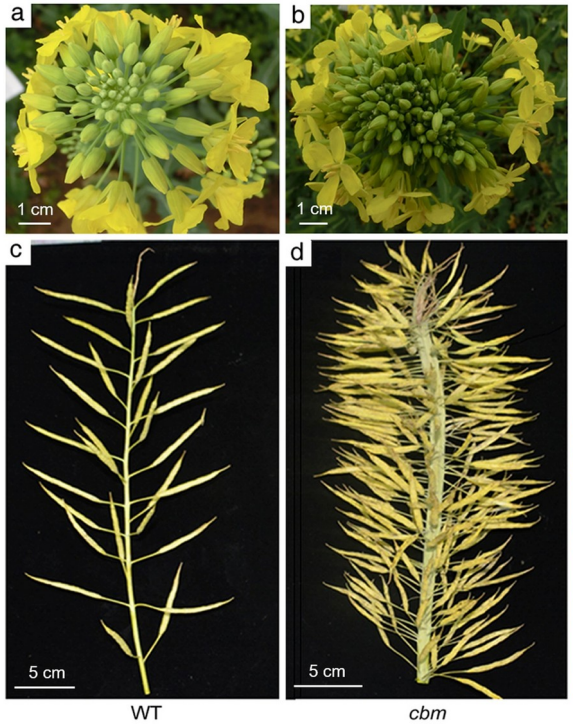
Bulked segregant analysis

- If you have two (in this case yeast) strains
  - One has phenotype of interest
  - One does not have phenotype of interest
- Perform mating with the strains
- Test ‘children’ or segregants for phenotype
- Group all positive segregants and all négatives in separate pools
  - Pool 1 is positive
  - Pool 2 is negative



Sequencing and thought process

- Mix segregants within each pools
  - Collect a tube containing the genomes of all the segregants of a single pool (typically 50 segregants per pool)
- Sequence the two genome mixtures
- Reads obtained will represent the whole pool
  - For SNVs unassociated with the phenotype should be 50% of positive reads have it and 50% of negatives have it
  - For SNV causal of phenotype should be in all 100% of positive reads and 0% of negative reads



Example two phenotypes rapeseed

Zhang, L., Cheng, X., Chen, D. *et al*. Integration of bulked segregant analysis and transcriptome sequencing reveals an interaction network associated with cluster buds trait in *Brassica napus*. *Theor Appl Genet* **138**, 200 (2025). https://doi.org/10.1007/s00122-025-04989-7

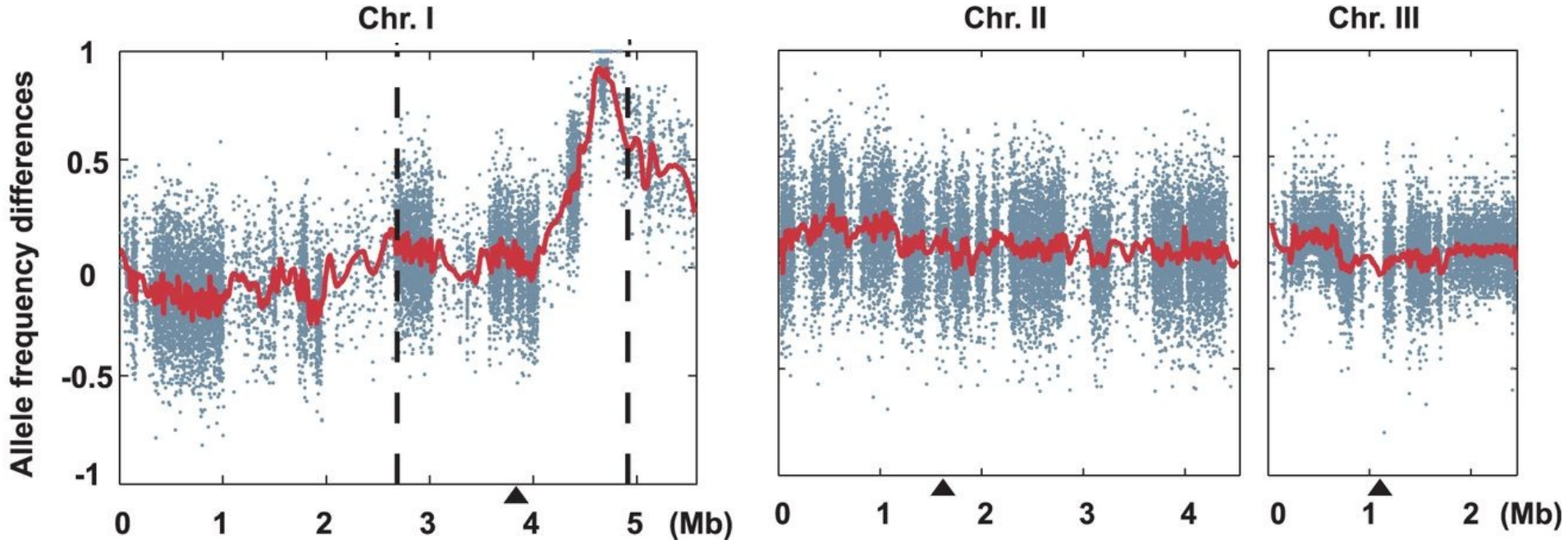
QTL-mapping

- Mapping where the “locus” is on the genome that cause the phenotype or “trait” is called quantitative trait loci mapping (QTL-mapping)
- After sequencing the two pools (positives and negatives) and having the reads of both positive and negative parent, one can do QTL-mapping
- Use program such as QTL-seq, see example
- Simple yet effective

```
qtlseq -r reference.fasta
-p parent.1.fastq.gz,parent.2.fastq.gz
-b1 bulk_1.1.fastq.gz,bulk_1.2.fastq.gz
-b2 bulk_2.1.fastq.gz,bulk_2.2.fastq.gz
-n1 20
-n2 20
-o example_dir
```

Pinpoint the important SNV!

- QTL-seq result
  - Important SNV is on Chr. I near the ~4.8 Mb mark.
    - Reduced SNVs from 10,000+ to smaller region on genome
  - Test causality!!



Summary

- Know your research question and hypothesis
- Raw reads
  - Make fasta using assembler (SPAdes) and annotation (AUGUSTUS)
  - Compare to existing genome using reference mapper (Bowtie2)

- Determine SNVs and impact
  - Variant caller like GATK – Haplotype caller
  - SnpEff or PLINK impact of SNVs
  - Bulk segregant analysis and QTL-seq

- Visualize using IGV or Manhattan plot



Q&A session

Always contact me:

Sjoerd.Seekles@unige.ch

SlideLizard

Questions?

Time left: -

Vote at: **attend.sl/w14r**

Votes: 0

Questions?

- (1)	animate (1)	blocking (1)	d3 (1)	from (1)	it (2)	on (1)	positioning (1)	simple.For (1)	than (1)
“important”: (1)	animations (1)	browser’s (1)	development.As (1)	Generator (1)	itself (1)	only (1)	possible (1)	size.This (1)	that (1)
a (3)	any (1)	can (2)	each (1)	GitHub (1)	layout (4)	open (1)	prevents (1)	slow (1)	the (8)
additional (1)	are (1)	Cloud (1)	even (1)	How (1)	license (1)	output (1)	quite (1)	source (1)	they (1)
algorithm (4)	as (3)	cloud.Note (1)	event (1)	hundred (1)	loop (1)	overlap (1)	recommended (1)	starting (1)	this (1)
always (1)	asynchronously, (1)	code (1)	few (1)	incredibly (1)	makes (1)	placed (1)	rendering (1)	step (2)	time (2)
an (1)	available (1)	configurable (1)	final (1)	into (1)	more (1)	placement (1)	requires (1)	stuttering.It (1)	to (2)
and (2)	be (2)	converting (1)	for (3)	is (4)	most (1)	placing (1)	run (1)	text (1)	under (1)

Preview